

## Experiment 3:

**Aim: To analyse the performance of a multi class confusion matrix by using choosen database with python code**

```
# Import libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
import matplotlib.pyplot as plt
import seaborn as sns

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Train the model
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)

# Display confusion matrix
plt.figure(figsize=(6,5))
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=iris.target_names,
            yticklabels=iris.target_names)

plt.title("Multi-Class Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.show()

# Print Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
# Print Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
```

